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Project Report on

“BANKING SYSTEM SIMULATION”

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Work carried out at



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2025-2026

BMS COLLEGE OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the OOPS with JAVA project titled “**Banking System Simulation**”
has been carried out by KHUSHI SHRIVASTAV (1BF24CS139) , KOMAL GUPTA
(1BF24CS144), MIMANSA SHARMA (1BF24CS173), NAINCY (1BF24CS186)
during the academic year 2025-2026.

Signature of the guide

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DECLARATION

We, KHUSHI SHRIVASTAV(1BF24CS139), KOMAL GUPTA (1BF24CS144), MIMANSA SHARMA(1BF24CS173), NAINCY (1BF24CS186), students of 3rd Semester, B.E, Department of Computer Science and Engineering, BMS College of Engineering, Bangalore, hereby declare that, this project work entitled "BANKING SYSTEM SIMULATION" has been carried out by us under the guidance of Prof. Tejaswini, Assistant Professor, Department of CSE, BMS College of Engineering, Bangalore during the academic semester Sep-Dec 2025. We also declare that to the best of our knowledge and belief, the project reported here is not from part of any other report by any other students.

Signature of the Candidates

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1.PROBLEM STATEMENT

Banking systems are an essential part of modern society, handling a large number of financial transactions and customer records on a daily basis. Traditional banking systems that rely on manual procedures for account management and transaction handling are often time-consuming and may lead to errors, data inconsistency, and inefficiency.

The problem addressed in this project is to design and develop a **simple Banking System Simulation using Java programming concepts**. The system should be able to simulate basic banking operations such as **account creation, user authentication (login), deposit of money, withdrawal of money, balance verification, and transaction history display**.

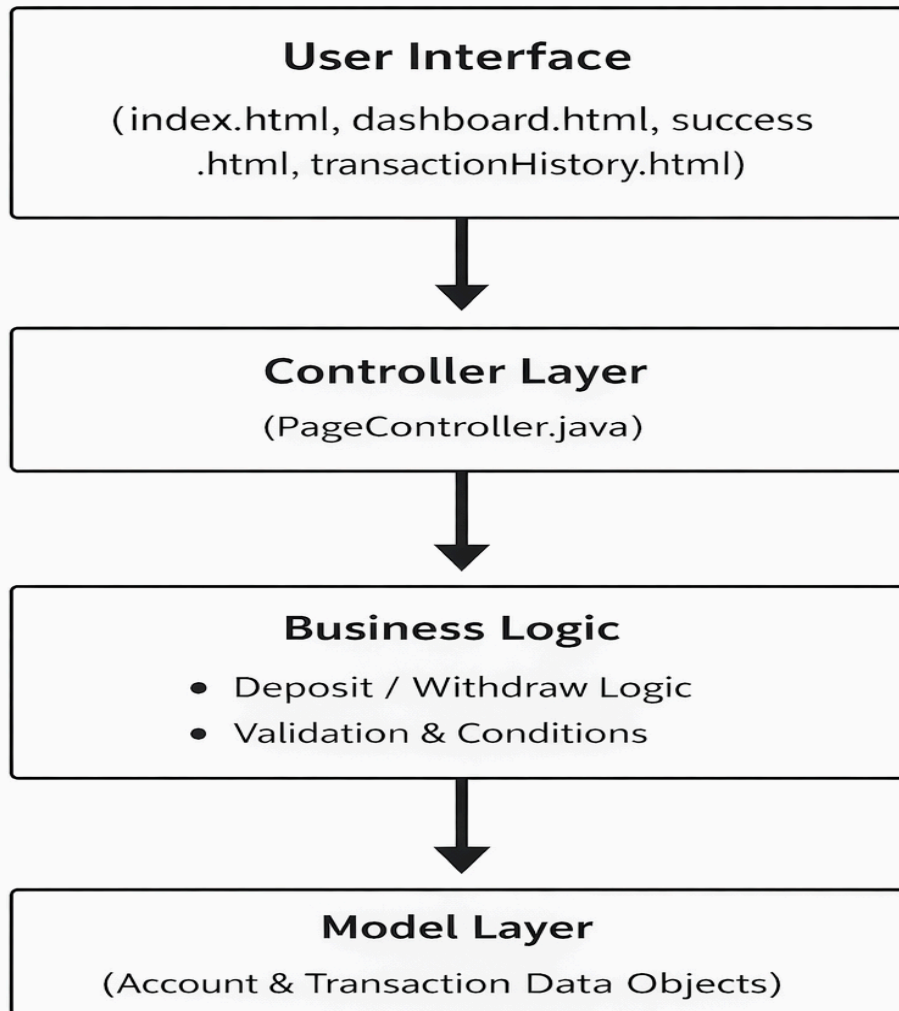
This project focuses on applying **core Java and object-oriented programming concepts** to create a structured and modular application. The goal is not to build a full-scale banking system, but to provide a clear understanding of how banking operations can be implemented using Java logic and MVC architecture in a simple and efficient manner.

2.INTRODUCTION

Banking applications are among the most commonly used real-world software systems. With the help of programming languages like Java, banking operations can be automated to improve efficiency, accuracy, and reliability.

This **Banking System Simulation** project is developed as a Java mini project to understand how real banking operations can be implemented using **core Java concepts and MVC architecture**. The project provides a simple and user-friendly web interface where users can create accounts, perform transactions, and view transaction history. The main emphasis of the project is on Java programming logic rather than advanced features.

3. OVERVIEW OF THE PROJECT (BLOCK DIAGRAM)



Block Diagram Explanation

The Banking System Simulation project follows a **layered** architecture.

4. TOOLS USED

The following tools and technologies were used for the development of the **Banking System Simulation** project:

Java

Java is used as the core programming language for developing this project. It provides platform independence, object-oriented features, and strong memory management. All business logic and backend processing are implemented using Java.

Spring Boot

Spring Boot is a Java framework used to build the web application. It simplifies Java web development by reducing configuration and providing embedded server support. Spring Boot is used to handle controllers, application flow, and MVC architecture.

Maven

Maven is used as a build and dependency management tool. It helps in managing external libraries and simplifies the build process of the project through the `pom.xml` file.

HTML & CSS

HTML is used to create the structure of web pages, while CSS is used for designing and styling the user interface. These technologies form the frontend of the application and allow users to interact with the system.

Embedded Tomcat Server

An embedded Tomcat server is used to run the Spring Boot application. It allows the project to be executed as a standalone web application without installing an external server.

Visual Studio Code (VS Code)

VS Code is used as the development environment for writing, editing, and managing the project code. It provides features like syntax highlighting, extensions, and integrated terminal support.

5.OOPS CONCEPTS USED & EXPLANATION

1. Classes and Objects

Classes are used to represent real-world entities like bank account and transactions. Objects are created to access the properties and methods of these classes.

2. Encapsulation

Account details such as balance and PIN are protected using access modifiers and accessed through methods, ensuring data security.

3. Packages

The project is organized into packages such as controller and test to maintain modularity and clean structure.

4. Abstraction

Only necessary functionalities such as deposit and withdrawal are exposed to the user, hiding internal implementation details.

5. Exception Handling

Conditions like insufficient balance are handled using conditional checks and validation logic to avoid system failure.

6. IMPLEMENTATION/ CODE

The project follows **MVC architecture**.

- **Controller Layer**

`PageController.java` handles page navigation such as home, dashboard, success, and transaction history. The controller also handles HTTP requests and redirects to appropriate views based on user actions.

- **View Layer**

HTML files like `index.html`, `dashboard.html`, `success.html`, and `transactionHistory.html` are used to display the user interface.

- **Configuration Files**

`application.properties` is used for application configuration such as server port.

- **Build Management**

`pom.xml` manages dependencies using Maven.

7. RESULT

The system successfully performs the following operations:

- Account creation with auto-generated account number
- Secure login using account number and PIN
- Deposit money into account
- Withdraw money with insufficient balance handling
- Display of transaction history

The image displays a user interface for a banking system, featuring two main sections: 'Create Account' and 'Login'.

Create Account

- Full Name
- 4-digit PIN
- Initial Balance
- Create Account

Login

- Account Number
- PIN
- Login

Account Created Successfully 🎉

Account Number	1003
Name	Shrishti
Balance	₹ 10000.0

[Back to Home](#)

[Logout](#)

Welcome, Shrishti

Account Number: 1003

Balance: ₹9000.0

Amount withdrawn successfully

Deposit Money

[Deposit](#)

Withdraw Money

[Withdraw](#)

[View Transaction History](#)

Transaction History	
Type	Amount
DEPOSIT	1000.0
WITHDRAW	2000.0
← Back to Dashboard	

Logout

Welcome, Shrishti

Account Number: 1003

Balance: ₹9000.0

Insufficient balance

Deposit Money

Enter amount

Deposit

Withdraw Money

Enter amount

Withdraw

View Transaction History

8.CONCLUSION AND FUTURE WORK

Conclusion

The Banking System Simulation project successfully demonstrates how basic banking operations can be implemented using Java. This project helped in understanding Java programming concepts, MVC architecture, and application flow.

Future Enhancements

- Database integration (MySQL)
- Secure authentication
- Fund transfer between accounts
- Admin panel
- Improved UI and validation

9. REFERENCES

1. Java Documentation – <https://docs.oracle.com/javase>
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4. GitHub – <https://github.com>

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